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Journal-3

Wednesday, September 17, 2025

- During class on September 17, we did our **presentation on the L293D H-Bridge Motor Driver IC**.
 - Learned how the IC is used to control the direction and speed of a motor.
 - Our teacher explained the new assignment about **PWM with Raspberry Pi**.
 - Me and my partner divided the work for this week.
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Thursday, September 18, 2025

- Read the assignment and understood the steps.
 - Learned that we need to test PWM with an **oscilloscope**.
 - Noted the duty cycle values (75% and 90%) to check later.
 - My partner checked the datasheet of the **L293D motor driver IC**.
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Friday, September 19, 2025

- Planned the **circuit design** for L293D on a breadboard.
 - Drew a simple sketch of the wiring.
 - Learned that both input pins should not be HIGH at the same time.
 - Wrote steps for safe wiring and setup.
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Saturday, September 20, 2025

- Worked on the **code** for motor driver control.
- Changed the program to use a **100 ms time period**.

- Added code so the **user can change motor direction**.
 - My partner checked the code in simulation.
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Sunday, September 21, 2025

- Studied how to check duty cycle (75% and 90%) on oscilloscope.
 - First, we tested PWM in **Multisim** and got waveforms for **75% and 90% duty cycles**.
 - Added comments in the code to show pin mapping.
 - My partner started writing the **README file** for GitHub.
 - Updated our notes with small corrections.
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Monday, September 22, 2025

- Set up the **breadboard with L293D** and checked connections.
 - Made sure to set both pins LOW before changing motor direction.
 - Partner updated the slides with circuit diagrams.
 - Uploaded draft code and notes on GitHub.
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Tuesday, September 23, 2025

- Tested PWM output on oscilloscope and saved images.
- Completed the README with steps to run the code.
- Uploaded all files, presentation, and this journal to GitHub and Conestoga.